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`makphot`: a set of Python routines to handle photometry

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ABSTRACT

My personal set of scripts for reducing publically accessible photometric time series data. Usually doing custom filtering for bad data and points with large errors.

1. ASASSN

Several cameras distributed around the globe. First check is that upper limits are given in the `Mag` column, so you can't read it as a float in Python. Filter those out, then filter out anything with 99.99 as mag or flux, as those are bad points.

Split by filter, since there are V band filters from early ASASSN work, then switch to g' later on.

Data is taken two or three times per night, not always the same camera, typically in a short burst.

Under the assumption we are not seeing real variability on these very short timescales, group these closely observed data points, then replace these points with a weighted average of the three original points, and calculate a new error bar.

The error bar is the larger of either (i) the STD of the two or three points or (ii) the mean of the ASASSN quoted errors. This is to avoid anomalously low error bars if the two points happen to be very close to each other in value but have very large reported error bars.

Filter out any nights with large dispersion.

(Luger et al. 2021).

REFERENCES

Luger, R., Bedell, M., Foreman-Mackey, D., et al. 2021, arXiv e-prints, arXiv:2110.06271. https://arxiv.org/abs/2110.06271